# Hyeonmin Jeon

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Applying for Ph.D., Fall 2027

## Education

**Yonsei University** Seoul, Korea
*M.S. in Industrial Engineering* Mar 2025 – Jun 2027 (expected)
Advisor: Prof. Kyong Joo Oh, Financial Investment AI Lab. GPA 3.97/4.0.
Relevant coursework: Stochastic Modelling Theory, Probabilistic Models & Bayesian Inference, Time-Series Analysis, Deep Learning Theory & Applications, Advanced Investment Engineering, Financial Engineering Topics, Corporate Finance.

**Chung-Ang University** Seoul, Korea
*B.S. in Electrical and Electronics Engineering* Mar 2019 – Feb 2025
GPA 3.59/4.0. Thesis: *Light-weighting Tiny YOLOv3: The Impact of Quantization on Real-World Applications* (advisor Prof. Minhyeok Lee).
Relevant coursework: Linear Algebra, Mathematics for Electrical Engineering, Numerical Analysis, Random Variables, Signals & Systems, Machine Learning, Artificial Intelligence, Algorithms.

## Research Interests

Bayesian learning and simulation optimization under input uncertainty; learned and multi-fidelity surrogate models; generative models (diffusion, normalizing flows) as simulation input models; sequential decision-making with statistical guarantees.

## Publications and Working Papers

Note. *: corresponding author.

### Journal Manuscripts

1. **Hyeon Min Jeon**, Hee Soo Lee*, Kyong Joo Oh*. Conditional Diffusion Transformers for Multi-Asset Scenario Generation and Portfolio Risk Measurement. *Expert Systems with Applications*, under review, 2026.
2. **Hyeon Min Jeon**, Hee Soo Lee*, Kyong Joo Oh*. FinPhasor: Phase-Preserving Measurement of Cross-Sectional Timing in Equity Markets. *Applied Soft Computing*, submitted, 2026.
3. Sang Hoe Kim, **Hyeon Min Jeon**, Dae Hyuk You, Jiyoung Jeon, Seungho Baek, Hee Soo Lee, Kyong Joo Oh*. Deep Convolutional Factor Prediction with Explainable AI for Enhanced Investment Strategy. *International Journal of Forecasting*, submitted, 2026.
4. Dae Hyuk You, Jiyoung Jeon, **Hyeon Min Jeon**, Sang Hoe Kim, Sang Hyuk Yoo, Tae Yoon Kim, Hee Soo Lee, Kyong Joo Oh*. Benchmark-Aware Enhanced Indexing Guided by the Volume–Price Index. *In preparation for resubmission*, 2026.

### Working Papers

5. **Hyeon Min Jeon**. Learned Surrogate Priors for Bayesian Model Selection: Density, Uncertainty, and the Cost of a Sampler. Lab seminar, January 2026. [Link]
6. **Hyeon Min Jeon**. Bayesian Capacity Sizing for an Exchange Matching Queue under Over-Dispersed Arrivals. 2026. [Link]

### Conference Presentations

7. YoungDoo Jeong*, HaeRim Kim, Dae Hyuk You, **Hyeon Min Jeon**, Andre de Castro, Taewoo Lee. A Study for Forecasting Solar Energy Power Generation. *45th International Symposium on Forecasting (ISF 2025)*, Beijing, July 2025. Talk, joint work with SAS Institute.
8. **Hyeon Min Jeon**, Minhyeok Lee. Dealing with the Imbalanced-Data Problem with a Diffusion Model. *KICS Fall Conference*, November 2023.
9. **Hyeon Min Jeon** et al. Personalized Sleep Score Prediction Model Using a Lifelog Dataset. *ETRI Human Understanding AI Paper Challenge*, 2024.

## Research Experience

**Financial Investment AI Lab, Yonsei University** — Seoul, Korea
*Graduate Research Assistant* — Mar 2025 – Present

- Generative scenario models for portfolio risk: a conditional diffusion transformer that generates joint multi-asset return scenarios, evaluated by the VaR/ES reserves set on its scenarios rather than by sample realism.
- Timing in the equity cross-section: forecasting complex wavelet scalograms so that the phase of price moves, and hence when each stock reprices, is preserved and used for cross-sectional ranking.
- Learned priors for Bayesian factor selection: replacing the mixture-of-$g$ prior with a normalizing flow that keeps an explicit density, checked against the exact posterior over models (TV distance 0.016) and tuned by Bayesian optimization; compared with a density-free GAN prior.
- Ranking and selection under input uncertainty: implemented and verified KN, feasibility determination, recycled observations and CRN, and studied how a fitted input model affects the probability-of-correct-selection guarantee.

**TIPS R&D Project with Urban Complex** — Seoul, Korea
*Researcher, artist-style generative engine (lab project)* — Mar 2025 – Aug 2026

- Built chain-of-thought style captioning with Qwen and trained a QLoRA style classifier on Qwen2-VL-7B (91.4% seen / 76.0% unseen; 96.0% fused with a DINOv2 prior).
- Trained SDXL-based generation with domain checkpoints routed by the classifier, and built the style memory module.

**GAIL (AI Lab), Chung-Ang University** — Seoul, Korea
*Undergraduate Researcher* — Dec 2022 – Dec 2024

- Diffusion-model oversampling for imbalanced fraud data; latent-diffusion portfolio study; lifelog sleep-score prediction; 3D point-cloud classification.

**Carbon-Neutral ESG Research Group, Chung-Ang University** — Seoul, Korea
*Team Lead, "Green-Navi" (Green AI track)* — Apr 2023 – Dec 2023

- Reinforcement learning for fuel-optimal urban routing in AIMSUN; presented at the research fair, November 2023 (MSIT/NRF Field-Linked Future-Leading Talent programme).

## Selected Projects

**Retina Exam Room Scheduling, with Nune Eye Hospital**
*Discrete-event simulation and reinforcement learning* — Feb 2026 – Present

- Recalibrated a discrete-event simulator to hospital records; compared rule, behaviour-cloned and PPO policies on a 44-day replay (time in room 12.2 → 8.2 min); read-only EMR integration and live comparison.

**Simulation-Methodology Reimplementations**
*rs-lab, Input-Uncertainty Select, TailFlow, Eco-Route RL, roughvol-lab* — Sep 2026

- Ranking & selection with verified guarantees; input-uncertainty effects on selection; conditional diffusion as a risk engine; $CO_2$-optimal routing on SUMO with P(correct selection) 0.956; rough volatility.

**Lucubra, research-agent platform**
*LLM agents for paper verification and referee reports* Sep 2026 – Present

**Light-weighting Tiny YOLOv3**
*Graduation thesis, team of three* 2024

- Int8 quantization with layer fusion; total footprint 1.6 GB → 100 MB, inference 22–64% faster across five devices, mAP −2.78 pt on PASCAL VOC.

## Teaching Experience

**Yonsei University** Seoul, Korea
*Teaching Assistant, Financial Engineering (undergraduate)* Spring 2026

## Awards and Honors

- Special Prize (data use), Chungcheong ICT Innovation Public-Data Competition 2024

## Military Service

**Mandatory military service**, Republic of Korea Mar 2020 – Sep 2021

## Skills

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|---|---|
| Programming | Python (NumPy, SciPy, pandas, PyTorch, CUDA), C++, MATLAB, TypeScript, SQL, LaTeX |
| Methods | Discrete-event and Monte Carlo simulation, ranking & selection, variance reduction, diffusion and flow models, Bayesian inference, reinforcement learning, time-series analysis |
| Tools | Eclipse SUMO, AIMSUN, SAS, Hugging Face, Linux multi-GPU servers |
| Languages | Korean (native), English (working proficiency) |

*Last updated on October 11, 2026.*